William Olson

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Education

University of Texas at San Antonio *Sept 2024 – Expected May 2028*
Bachelor of Science in Physics – Minors in Mathematics and Astrophysics

- Cumulative GPA: **4.00/4.00**
- Relevant Coursework: Classical Mechanics, Mathematical Physics, Differential Equations, Linear Algebra

Honors & Awards

Distinguished Presidential Scholarship *2024 – Present*
- Competitive full academic scholarship awarded by the University of Texas at San Antonio.

Texas Excellence Grant *2024 – Present*
- Merit-based financial award recognizing academic achievement.

University Honors College *2024 – Present*
- Selected member of the UTSA Honors College.

NSF REU Participant – Breakthrough Listen Initiative *Summer 2026*
- Selected to conduct summer research with the Breakthrough Listen Initiative at the University of California, Berkeley.

Research Experience

NSF REU Student Researcher – Breakthrough Listen Initiative *Berkeley, CA*
June 2026 – Present

- Conduct radio technosignature searches using Green Bank Telescope observations as part of the Breakthrough Listen collaboration.
- Evaluate the performance of turboSETI and BLISS across narrowband, mid-resolution, and rawspec datasets using known spacecraft transmissions.
- Develop Python analysis pipelines to identify, visualize, and compare candidate signals while investigating the effects of spectral resolution on detection sensitivity.
- Analyze algorithm performance and candidate validation workflows to understand limitations of automated technosignature searches.

Machine Learning for Ionization Cone Detection in Active Galactic Nuclei *San Antonio, TX*
Sept 2024 – Present

- Develop U-Net deep-learning models for automated identification and segmentation of ionization cones in active galactic nuclei.
- Design a modular Python framework for synthetic JWST image generation, augmentation, model training, evaluation, and inference.
- Apply trained models to JWST NIRCam and MIRI observations to investigate generalization between simulated and observational datasets.
- Collaborate on physically motivated machine-learning methods for astronomical morphology studies.

AGN Torus and Polar Wind Modeling *San Antonio, TX*
Sept 2024 – Present

- Model infrared emission from AGN dust structures using `FortesFit` with CAT3D, CAT3D-Wind, and CLUMPY radiative transfer models.
- Perform forward modeling and image deconvolution to investigate dust geometry and polar emission.
- Compare theoretical model predictions with JWST observations to support interpretation of machine-learning analyses.

Research Interests

Active Galactic Nuclei (AGN); galaxy evolution; astronomical machine learning; scientific artificial intelligence; radio astronomy; technosignature searches; JWST data analysis; astronomical instrumentation

Open-Source Software & Programs

Ionization Cone Automation Pipeline *https://github.com/2xwlll/IonizationCon Automation*

- Developed a modular Python framework for synthetic AGN image generation, machine-learning training, inference, and evaluation.
- Implemented reusable workflows for astronomical image processing, data augmentation, and performance assessment.
- Designed to support reproducible deep-learning experiments on JWST observations.

Honors College Event Analyzer *https://github.com/2xwlll/SPICESsense*

- Developed a Python application that analyzes attendance trends across Honors College events.
- Performs statistical analysis using event categories, SPICES themes, and keyword relationships.
- Generates visualizations to support data-driven planning and assessment of student engagement.

Presentations

Olson, W., Schneider, J., Zhang, L., Packham, C.
A Physically Motivated Machine Learning Framework for Systematic Identification of AGN Ionization Cones.
Poster presented at the Kyoto Torus Workshop, Kyoto, Japan, May 2026. **(International)**

Ortuno Lopez, L., **Olson, W.**, Schneider, J., Zhang, L., Gutierrez, J. B., Packham, C.
Using Artificial Intelligence to Leverage JWST Datasets to Explore Active Galactic Nuclei.
Poster presented at the Cosmic Horizons Conference, Austin, Texas, May 2025. **(National)**

Olson, W., Schneider, J., Zhang, L., Packham, C.
Physics-Guided Machine Learning for Automated Tracing of AGN Ionization Cones.
Poster presented at the UTSA - SwRI Symposium, San Antonio, Texas, April 2026. **(Community)**

Olson, W., Schneider, J., Zhang, L., Packham, C.
A Physically Motivated Machine Learning Framework for Systematic Identification of AGN Ionization Cones.
Poster presented at the UTSA Spring Symposium, San Antonio, Texas, April 2026. **(University)**

Olson, W., Schneider, J., et al.
LIRA: Lunar Infrared Array Mission Concept.
Oral presentation for the UTSA Leadership and Science Integration Course, May 2025. **(University)**

Olson, W., Jones, M.
Water Quality Assessment of Lake Boerne.
Poster presented at the APPEX Student Research Symposium, San Antonio, Texas, October 2024. **(University)**

Programs & Training

Academic Personal and Professional Excellence (APPEX) *Aug 2024 – Present*

- Selected for a competitive Ph.D. preparation program supporting undergraduate researchers pursuing graduate education.
- Completed training in scientific communication, research ethics, graduate fellowship preparation, and professional development.
- Participated in workshops on NSF Graduate Research Fellowship Program (GRFP) applications, graduate school preparation, and research presentation skills.
- Eligible for competitive conference travel funding through the APPEX program.

College of Science Work Study Research Training Program (COS WS-RTP) *Jan 2026 – Present*

- Selected for a competitive undergraduate research training program emphasizing laboratory readiness and

professional development.

- Completed training in responsible conduct of research, laboratory safety, scientific communication, and graduate school preparation.
- Participated in seminars connecting undergraduate researchers with faculty mentors and research opportunities across the College of Science.

Leadership & Outreach

- Volunteer track and field coach, mentoring student athletes in sprint training, technique development, and athletic preparation.
- Engage in STEM outreach, giving speeches on College and the Ph.D. pathway for James Madison Highschool.

Affiliations

- UTSA Mid-Infrared Astronomy & Exoplanet (MAE) Research Group – Student Researcher
- Breakthrough Listen Collaboration – NSF REU Participant
- Society of Physics Students (SPS), University of Texas at San Antonio
- Academic Personal and Professional Excellence (APPEX) Ph.D. Training Program
- College of Science Work Study Research Training Program (WSRTP)
- Honors College Advisory Board (HSCAB) (past)

Technical Skills

Programming Languages: Python, Bash

Scientific Computing: NumPy, SciPy, Pandas, Astropy, GNU Radio Signal Processing

Machine Learning: PyTorch, U-Net, segmentation_models_pytorch

Astronomy Software: MAST Archive, SAOImage DS9, FITS

Radio Astronomy: TurboSETI, BLISS, Blimpy, Filterbank Analysis, Spectral Analysis

Visualization: Matplotlib

Development Tools: Git, Linux, Jupyter Notebook, LaTeX